How incremental are predictions? Insights from cognitive modeling and eye-tracking

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Visual world studies have shown that speech processing, particularly word recognition, is highly incremental, with listeners using phonological information as the speech signal unfolds[1]. In contrast, computational models typically assume that predictions are generated only when a word has fully unfolded and all its internal information is available[2,3,4] and studies on prediction typically treat words as the relevant units for prediction update[5]. Crucially, sub-lexical cues may allow listeners to generate predictions incrementally as each morpheme is processed. This raises a key question: Do listeners generate predictions based on morpheme-level information as it becomes available, or do they wait until the whole word is processed? To test this, we implemented two cue-based retrieval models of prediction—one operating at the word level, the other at the morpheme level—and evaluated their predictions in two visual-world eye-tracking experiments.

Cognitive modeling. We implemented two cue-based retrieval models[6]. Both models share the same underlying assumption: predictions are cue-based memory retrievals[2]. The models differed only at the level of granularity for predictions: word-level[2] versus morpheme-level[7]. Cue matching was binary for all cues (gender, color, etc.) and graded only for semantics.

Visual world experiments. We tested German native speakers (n = 62, data collection ongoing) in two experiments that examined predictions based on German adjectives, which consist of at least two morphemes (e.g., blau-er *blue*$_{.MASC\text{-}NOM}$) that provide distinct cues: a semantically marked stem cue and a gender-marked suffix cue. In **Experiment 1**, participants viewed two-object displays while hearing German instructions like “There is a blue balloon!” (Fig. 1D). The target object matched the information from the adjective stem (color cue) and suffix (gender cue). The color cue could be evaluated visually. Interference was manipulated by varying the competitor image: unambiguous (mismatching both cues), gender-interference (matching the suffix cue only), and color-interference (matching the stem cue only). In **Experiment 2**, participants viewed greyscale displays while hearing instructions like “There is a hot coffee!” (Fig. 2D). Here, the semantic cue from the adjective stem could not be verified through visual information and thus depended solely on the strength of the association between the adjective and the objects. Interference was manipulated via this association: in the low interference condition, the adjective stem was strongly associated with the target but not with the competitor (‘hot coffee’ vs. ‘hot ice cream’); in the high interference condition, the adjective was changed to favor the competitor (‘cold ice cream’), which was strongly associated with the competitor but only weakly with the target (a rating experiment confirmed the strength of the association).

Results and discussion. In Exp. 1, the word-level computational model predicted similar prediction onsets across conditions, whereas the morpheme-level model predicted a delayed onset in the color competitor condition. This is due to the morpheme-level model allowing activation to spread already after the stem; in the color-interference condition, this results in equal activation of both objects. Eye-tracking data support the morpheme-level model: the onset of the predictive effect occurred later in the color-interference condition than in the unambiguous and gender-interference conditions, indicating sensitivity to morpheme-level cues (Fig. 1A–C). In Exp. 2, the word-level model predicted a delayed prediction in the high interference condition. The morpheme-level model also predicted this delay but differed by predicting a temporary preference for the competitor before the target was predicted. This occurred because spreading activation during stem processing favored the competitor, whose semantic cues were more strongly associated with it than with the target. Eye-tracking data supported the morpheme-level model: fixations in the high-interference condition showed a delayed onset and a temporary competitor-over-target preference (Fig. 2A–C). Overall, these results support the view that predictive processing operates at the sub-lexical level. This shows that predictions are more incremental than often assumed and highlights the relevance of sub-lexical predictions especially in morphologically rich languages like German, and the need for theories to reflect this granularity.

Figure 1. Experiment 1: (A–B) Model-predicted fixation probabilities from [3] and [6]; (C) empirical fixation curves with 95% CI; (D) experimental design.

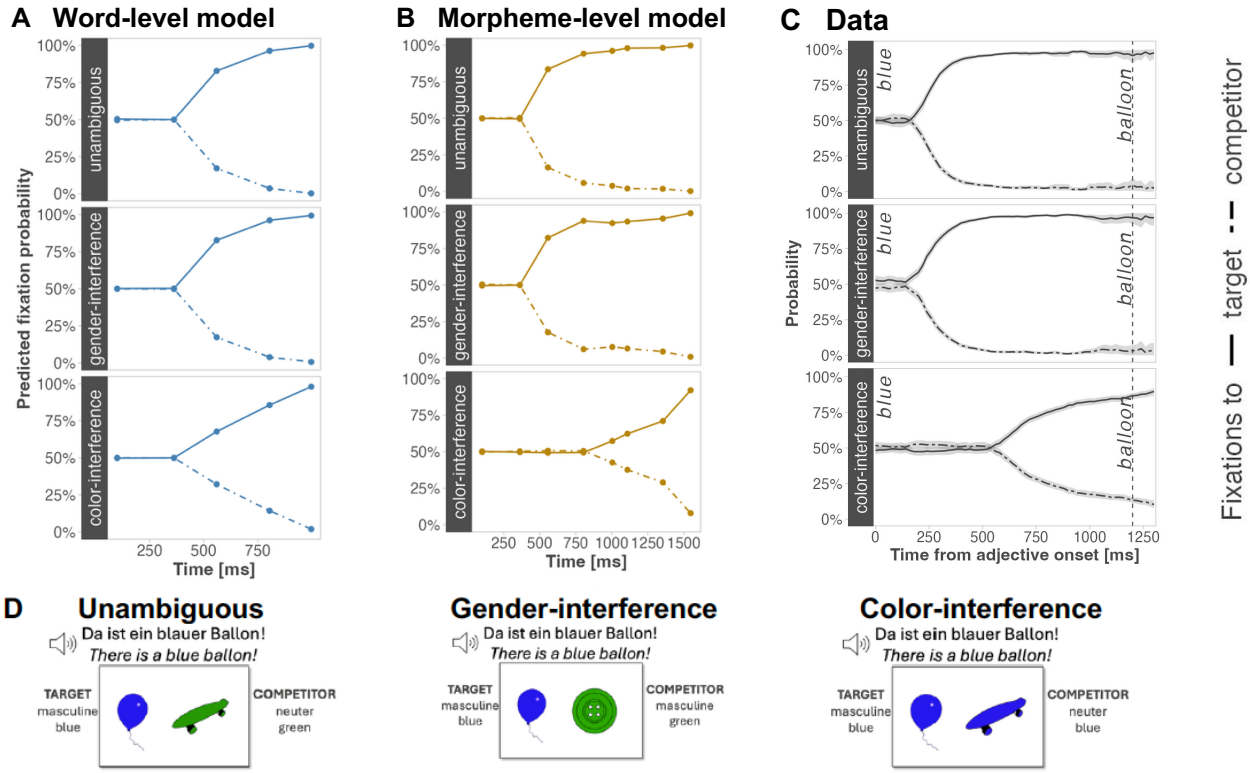
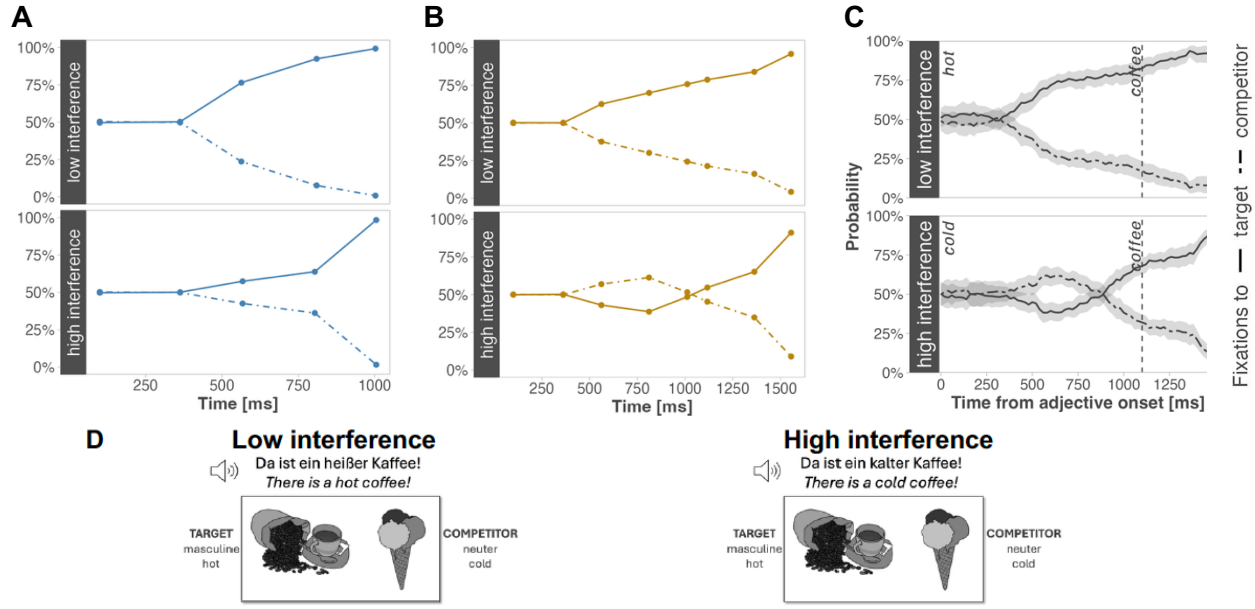


Figure 2. Experiment 2: Same structure as Figure 1, showing model predictions, empirical fixations, and the experimental design.



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